LESSON PLAN

NDT (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Lubrication system main components and its purposes, operation of each components of system, Air cooling and sealing of engine main bearings
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Lubrication system main components and its purposes, operation of each components of system, Air cooling and sealing of engine main bearings
5. Specific Objectives	20.1.2.1 Trainee will be able to understand and appreciate noise separation methods, thrust reversal mechanism, after burning fuel system, lubrication system, thrust augmentation devices, cooling and thrust calculations

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Lubrication System Diagrams, Safety Precautions regarding Hot Oil, Instructor Supervision of Component Identification
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the engine lubrication system?
(b) Name two main types of lubrication systems used in aircraft engines.
(c) Why is oil cooling necessary for high-performance engines?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Oil Pump Operation and Pressure Generation • Oil Filter Function and Bypass Mechanism • Oil Cooler Air Cooling Principles • Main Bearing Sealing Methods • Scavenge Pump Operation	L & Ds	PPT, IB	15 min
(b)	• Detailed Operation of Oil Pressure Regulator • Air Cooling System Airflow Dynamics • Seal Integrity and Leakage Prevention • Lubrication of Main Bearings • System Pressure Monitoring	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting Lubrication System Failures • Maintenance of Oil Coolers and Filters • Inspection of Main Bearing Seals • Oil Analysis and Contamination Checks • System Performance Verification	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the air cooling system removes heat from the oil.
(b) Describe the role of the scavenge pump in the lubrication circuit.
(c) What are the consequences of failed main bearing seals?

12. Home Assignment and Follow Up	Prepare a brief note on the calculation parameters for engine thrust and cooling requirements to be discussed in LP 13.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Ice protection, hot air protection, electrical system Calculation of engine thrust, various parameters and tests, adjustments of engine controls	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min